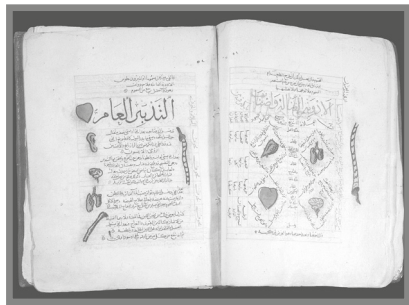


algebraic equations. Crucially, the Hindu mathematicians also employed a decimal notation, which one of the members of the House of Wisdom, Al-Khwarizmi (c. 780–830), adopted and described in *The Book of Addition and Subtraction According to Hindu Calculation*.

Furthermore, Al-Khwarizmi also explained a method of calculating the square roots of numbers, and pioneered work on algebraic equations. He and his fellow scholars made rapid strides in geometry, taking as their starting point Euclid's and Archimedes's work on spheres and cylinders.

Astronomy and medicine

Al-Khwarizmi compiled the first known tables of daily prayer times at Baghdad, his calculations assisted by direct astronomical observation. The early Islamic astronomers drew from Ptolemy's *Almagest*, adopting his view that the Earth was at the center of the solar system, and that the planets rotated around it along the lines of eight spheres. They also learned from Hindu astronomers, translating and perfecting Indian *zij*, or tables of planetary positions, and continued to refine Ptolemy's system, only occasionally (as in the work of the 10th-century astronomer al-Biruni) toying with a heliocentric system that had the sun at its center. Their calculations were made simpler when in the mid-eighth century they adopted the astrolabe, an instrument in which the celestial sphere was projected onto a flat plane marked with latitude and longitude lines.



The Canon of Medicine by ibn Sina or Avicenna (980–1037) set the standard for medicine in the Islamic world and medieval Europe, and remained an authority for centuries.

By the 13th century, Islamic astronomy was at its zenith, and in 1259 a great observatory was constructed at Maragha in eastern Iran. Here Nasir al-Din al-Tusi and his successors made fine adjustments to account for slight discrepancies in the orbit of the planets, assisted by mechanical clocks that enabled them to record their observations in fine detail. Muslim scholars made advances in many other areas, too, first building on the base of Greek manuscripts translated into Arabic, and then making their own discoveries. They did not accept the theories of the ancients uncritically: al-Haytham (died 1039) produced a key work, the *Book of Optics*, in which he speculated that sight was the result of light traveling from an object to the eye, rather than the other way around as Ptolemy had theorized. Arab physicians continued to make progress, combining their practical observations with theoretical analysis. Al-Razi (died 925) produced the first description of smallpox and measles, as well as compiling a medical compendium that began a tradition of such encyclopedias, culminating in the *Canon of Medicine* by ibn Sina (who was known as Avicenna in the West). Composed around 1015, it included separate sections for diseases that are specific to one body part, and those that afflict the body as a whole.

Islamic science spreads

The Islamic expansion that began in the mid-7th century not only absorbed ancient centers of learning such as Alexandria, but also brought the Muslim world to the fringes of Western Europe through the conquest of Spain (from 711) and Sicily (from 827). A tradition of Islamic learning embedded itself in both areas, and particularly in the Iberian Peninsula, known to the Arabs as al-Andalus. The court established there in 756 by Abd al-Rahman I, a refugee Umayyad prince who had escaped the Abbasid revolution, became a magnet for scholars from the East, and its libraries became a repository of precious ancient texts that had been translated into Arabic. »